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The Mean Wave, Wind and Current Forces on Offshore Structures and Their Role in the Design of Mooring Systems

By

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Abstract

The mean forces induced by waves, wind and current on tankers, barges and other structures, are very important for the selection of an appropriate passive or active positioning system. In the present paper tools are given for the determination of these forces, especially for tanker shaped bodies, taking into account the main influence of water depth. The way in which the estimation of the mean forces has to be used for a preliminary design of a mooring system will be dealt with.

Introduction

The problem of designing an adequate mooring or anchoring system for floating offshore structures is rather complicated not at least due to the non-linear nature

of the problem. It is a problem of dynamics and the moored structure may be regarded as a mass-spring system. The total load, exerted by the environment, consists of the oscillating wave forces and of forces, which vary at a frequency much lower than the wave frequency: the wind, current and wave drift forces. When designing an anchor system, three aspects of these very slowly varying forces are of importance.

1. As will be illustrated at the end of this paper the almost steady wind, current and wave drift forces are most significant for the final determination of the diameter of the anchor lines. This is due to the fact that the main purpose of an anchoring system is to hold the structure on an average position. Basically the force required for this station keeping equals the steady or the slowly

oscillating wave, wind and current forces on the structure.

2. The almost steady wind, current and wave drift forces cause a shift of the neutral position of the moored object, around which the oscillation due to waves occurs. Since the relationship between the horizontal displacement of the anchored structure and the restoring force of the anchor system is almost always non-linear, the shift of the neutral position causes a change in the spring rate, and consequently in the dynamic response. However, generally the natural frequency of the horizontal motion of the anchored structure is considerably smaller than the wave frequency, also after an increase of the spring constant, due to the mean shift of the structure, but this has to be checked for each special design. Therefore it is very important to study this effect of the wind, current and wave drift forces in an early stage of the design, to be sure that resonance at the wave frequency will be avoided.

3. Since the wind, current and wave drift forces are slowly fluctuating forces, the risk exists that resonance occurs at the very low frequencies, at which these forces oscillate. Low frequency oscillations in the horizontal plane have been observed in model experiments as well as at full scale. The nature of this phenomenon is not yet fully understood, but in general it can be attributed to the slowly varying wave drift force as a result of the varying wave height in irregular seas. Attention has to be paid to the dynamic forces due to wind gusts,

while the variation in the current velocity occurs at a much too low frequency to be of importance.

The first two aspects discussed here, have to be taken into account during the preliminary design, and will be of significant importance for this design. For this aspect the wind, current and wave drift may be regarded as steady phenomena, inducing only constant forces. For the investigation of the dynamic character of especially the wave drift force and its effect on the dynamic response of the anchored object, model tests are still indispensable.

Figure 1 shows a flow diagram, depicting a possible design process for an anchor system. In general, the process will have to be repeated for a number of different conditions. For a drill rig, for instance the anchor system must be strong enough to withstand the extreme loads in the survival condition, while for the maximum operational wave condition the motion of the rig may not exceed a certain value. In a certain condition, it may be necessary to consider various combinations of wave, current and wind directions.

In this paper we will deal with the determination of the steady forces, mainly on tanker hulls. The nature of the wind, current and wave drift forces will be briefly discussed and data will be given for an estimate of these forces. The problem of the choice of the design environmental conditions will be left out of consideration. An example how to use these data for the design of an anchor system will be given.

Description of steady forces and moments

Wind forces and current forces are frequently presented as being composed of a drag force, in the direction of the flow, and a lift force, perpendicular to the flow direction. However, when describing the dynamic behaviour of a floating object, the equations of motion are usually related to a structure bound system of coordinates, with its origin in the midship section of the structure. Therefore it is convenient to describe the steady forces as a longitudinal and a transverse component applying in the midship section and a moment around the vertical axis.

Figure 2 shows a sketch in which the sign of the forces and moment, and the direction of wind, waves or current are defined.

The force due to wind

Like all environmental phenomena, wind has a stochastic nature which greatly depends on time and location. It is usually characterized by fairly large fluctuations in velocity and direction. It is common meteorological practice to give the wind velocity in terms of the average over a certain interval of time, varying from 1 to 60 minutes or more. The variation in mean velocity is very slow compared with the wave period. The fluctuations around the mean value will impose dynamic forces on the structure, but in general these aerodynamic forces may be neglected in comparison with the hydrodynamic forces, when considering the dynamic behaviour of the floating body.

Therefore, we will consider the wind velocity as a steady value, both in magni-

tude and direction, resulting in constant forces and a constant moment acting on the structure.

The role which wind plays in the determination of the environmental conditions of floating structures is twofold:

- On the part of the structure exposed to the air, wind forces are exerted due to the stream of air around the various parts. For the determination of these forces information is required about the local winds only.
- The forces exerted by the wind on the water surface cause disturbances of the still water level, thus generating waves and current, which induce forces on the submerged part of the structure. To determine this effect of the wind, information is required about the wind and storm conditions in a much larger area.

The wave and current generating character of the wind will be left out of consideration. The effects of waves and currents will be dealt with separately.

Local winds are generally defined in terms of the average velocity and average direction related to a certain height above the still water level. The standard height above the water surface for which generally the wind velocity data are given amounts to 10 metres or 30 feet. A number of empirical and theoretical formulas is available in literature to determine the wind velocity at other heights. An adequate vertical distribution of the wind speed is represented by (see ref. (1)):

$$\frac{V(Z)}{V(10)} = \left(\frac{Z}{10}\right)^{1/7}$$

in which

$V(Z)$ = wind speed at height Z above the water surface

$V(10)$ = wind speed at 10 metres height above the water surface

The total force and moment experienced by an object exposed to wind, is partly of viscous origin (pressure drag) and partly due to potential effects (lift force and moment). For blunt bodies, the wind force may be regarded as independent of the Reynolds number and proportional to the square of the wind velocity.

Wind load data for tankers

The forces and moment exerted by wind on tankers can be calculated from:

$$X_w = \frac{1}{2} \rho_a V_w^2 C_{xw}(\alpha) \cdot A_T$$

$$Y_w = \frac{1}{2} \rho_a V_w^2 C_{yw}(\alpha) \cdot A_L$$

$$N_w = \frac{1}{2} \rho_a V_w^2 C_{nw}(\alpha) \cdot A_L \cdot L$$

in which

X_w = steady longitudinal wind force

Y_w = steady transverse wind force

N_w = steady yaw wind moment

ρ_a = density of air

V_w = wind velocity

α = wind direction

A_T = exposed transverse area

A_L = exposed lateral area

L = length of the ship

C_{xw} , C_{yw} and C_{nw} are coefficients, depending on the angle of incidence of the wind.

In literature the results of wind tunnel experiments are given for various types of vessels. From several papers the wind data on tanker hulls were collected, and the force and moment coefficients C_{xw} , C_{yw} and C_{nw} were expanded in Fourier series as a function of the angle of incidence:

$$C_{xw} = a_0 + \sum_{n=1}^{\infty} a_n \cos n\alpha$$

$$C_{yw} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

$$C_{nw} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

From the harmonic analysis it was found that a fifth order representation of the wind data is sufficiently accurate for preliminary design purposes.

In Table II through IV the Fourier coefficients are given for the longitudinal and transverse force, and the yawing moment, for a number of tankers. Particulars of the tankers are listed in Table I, where also the reference numbers are given of the publications, from which the data were taken. Figure 3 shows, as an example, the measured wind forces and moment together with the Fourier approximation, for one of the tankers.

The wind load on other structures

The wind forces and moments on other types of structures, as for instance floating semi-submersible platforms, can be approximated by dividing the structure in a number of components, all with a more or less elementary geometry, and estimating the wind force on each element. For a lot of simple geometrical forms such as spheres, flat plates and cylinders of various cross sectional shapes, the drag and lift coefficients are given in literature. Reference is made of the publications of Hoerner (6) and Delany and Sorensen (7). The total wind load on the structure is then found by adding the contributions of all the component parts.

The force due to current

There are several independent phenomena responsible for the occurrence of current: the ocean circulation system, resulting in a steady current, the cyclical change in lunar and solar gravity, causing tidal currents, wind and differences in density. The steady velocity at the water surface due to wind amounts to about 3 percent of the wind velocity (at 30 ft height). Tidal currents are of main importance in areas of restricted water depth and can attain values up to 10 knots. However, these extreme velocities are rare. A 2 or 3 knots tidal current speed is common. The prediction of the magnitude of tidal currents is a special science. Some predictors take not less than 60 parameters into account in their prediction procedure.

Although for floating structures the surface currents will be the governing ones, the vertical current distribution may also be of importance, especially for the case of restricted water depth. For the design of an anchor system of a floating structure the designer is especially interested in the probability that a particular extreme current velocity will be exceeded during a certain period of time. Observations obtained from current speed measurements are indispensable for that purpose. It may be usefull to split up the total measured current in two or more components, for instance in a tidal and a non-tidal component, since the direction of the various components will be different, in general. The variation in velocity and direction of the current is very slow, and current may therefore be considered

as a steady phenomenon.

The forces and moment exerted by current on a floating object is composed of the following parts:

- A viscous part, due to friction between the structure and the fluid, and due to pressure drag. For blunt bodies the frictional force may be neglected, since it is small compared to the viscous pressure drag.
- A potential part, with a component due to a circulation around the object, and an other component due to the free water surface (wave resistance). In most cases, the latter component is small in comparison with the first.

Current load data for tankers

The forces and moment exerted by current on a tanker hull can be described by:

$$\begin{aligned} X_c &= \frac{1}{2} \rho_w V_c^2 C_{xc}(\alpha) \cdot A_{TS} \\ Y_c &= \frac{1}{2} \rho_w V_c^2 C_{yc}(\alpha) A_{LS} \\ N_c &= \frac{1}{2} \rho_w V_c^2 C_{nc}(\alpha) A_{LS} \cdot L \end{aligned}$$

in which

- X_c = steady longitudinal current force
- Y_c = steady transverse current force
- N_c = steady yaw current moment
- ρ_w = density of water
- V_c = current velocity
- α = angle of incidence
- A_{TS} = submerged transverse area $= \beta \times T$
- A_{LS} = submerged lateral area $= L \times T$
- L = length of the ship
- T = draft of the ship
- C_{xc} , C_{yc} and C_{nc} are coefficients, depending on the current direction.

At the N.S.M.B., the current loads have been measured on several tanker models of different size. From the results the

coefficients C_{xc} , C_{yc} and C_{nc} were calculated. For flow in the longitudinal direction a tanker hull is a rather slender body, and consequently consists the longitudinal force mainly of frictional resistance. The total longitudinal force was very small for the relatively low current speed, and could therefore not be measured very accurately. Moreover, extrapolation to full scale dimensions is difficult, since the longitudinal force is affected by scale effect. For mooring problems, the longitudinal current force will hardly be of importance. An estimate of its magnitude can be made by calculating the flat plate frictional resistance:

$$X_c = \left(\frac{0.075}{(\log R_n - 2)^2} \right) \cdot \frac{1}{2} \rho_w S V_c^2 \cos \alpha \cos \alpha$$

$\Delta \log \theta \text{ (fig. 8)}$

in which

$$R_n = \text{the Reynolds number} = \frac{V_c \cos \alpha L}{\nu}$$

ν = kinematic viscosity of water

S = the wetted surface

Extrapolation of the transverse force and yaw moment to prototype values is no problem. For flow in the transverse direction the tanker is a blunt body, and since the bilge radius is small, flow separation occurs in the model in the same way as in the prototype. Therefore, the transverse force coefficient and the yaw moment coefficient are independent of the Reynolds number.

The coefficients for the transverse force and the yaw moment were expanded in a Fourier series, as was done for the wind load coefficients:

$$C_{yc} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

$$C_{nc} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

The average value of the Fourier coefficients for the fifth order Fourier series, are given in Table V. These results apply to deep water. For shallow water, the current force and moment coefficients have to be multiplied by a coefficient, which is given in Figure 4 on a base of the water depth-draft ratio. In the data, given in Table V, the influence of the free water surface is included. This influence, however, depends on the water depth and on the Froude number, and consequently changes if the current velocity or the tanker dimensions change. For the condition to which these data apply, deep water and a prototype current speed in the order of 3 knots, the effect of the free water surface is very small. For a case of a small underkeel clearance and a current direction of 90 degrees damming up of the water at the weather-side and a lowering of the water level at the lee side of the ship occurs.

The current load on other structures

The current load on other types of floating structures can be estimated in the same way as was described for the wind load in a previous section.

Wave drift forces

A structure floating in waves experiences forces and moments which can be determined if the velocity potential ϕ of the water motion around the structure is known. By integrating the component of the pressure in a particular direction over the hull of the structure, the force component in this direction can be calculated. The pressure can be obtained

from the Bernoulli equation for non-steady flow:

$$p = p_0 + \rho_w \frac{\partial \phi}{\partial t} - \rho_w g z + \frac{1}{2} \rho_w V^2$$

in which

p = pressure

p_0 = atmospheric pressure

ϕ = velocity potential

g = acceleration of gravity

V = velocity of water motion

$$V^2 = \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2 + \left(\frac{\partial \phi}{\partial z}\right)^2$$

In the theory of periodic ship's motion in waves the velocity term $\frac{1}{2} \rho_w V^2$ is neglected, since it has only a second order influence on the oscillatory behaviour. However, it is this term which is responsible for the steady drift force. Representing the velocity potential by a periodic function proportional to the amplitude ζ_a of the incident wave

$$\phi = \phi'_a \cdot \zeta_a \cdot \sin \omega t$$

in which

ω = frequency of waves

it follows that

$$\frac{1}{2} \rho_w V^2 = \frac{1}{2} \rho_w \cdot \zeta_a^2 \cdot \sin^2 \omega t \left\{ \left(\frac{\partial \phi'_a}{\partial x}\right)^2 + \left(\frac{\partial \phi'_a}{\partial y}\right)^2 + \left(\frac{\partial \phi'_a}{\partial z}\right)^2 \right\}$$

By taking the average value of the pressure during one period ($T = \frac{2\pi}{\omega}$) it will be clear that all periodic terms vanish and only the velocity term $\frac{1}{2} \rho_w V^2$ gives a contribution. Consequently the steady drift force on a structure in waves is proportional to the square of the height of the incident wave. Maruo (8) shows that the lateral drift force Y'_d per unit length on an infinitely long cylinder

(2-dimensional case; no diffraction) in beam seas satisfies:

$$Y'_d = \frac{1}{2} \rho_w g \zeta_{ar}^2$$

in which

ζ_{ar} = amplitude of wave reflected and scattered by the body

He also indicates that the amplitude of this wave is proportional to the relative motion between the oscillating cylinder and the wave.

Wave drift load data for tankers

Ogawa (9) applied Maruo's theory on a captive series 60 model (block coefficient 0.70) in beam and bow quatering waves and he shows that

$$Y'_d = \frac{1}{2} \rho_w g \zeta_{ar}^2 \cdot \sin^2 \alpha$$

in which

α = direction of waves relative to ship

He calculated the amplitude of the reflected and scattered wave using the strip theory for two wave directions (90° and 120°) relative to the captive vessel and found a reasonable agreement with the measured results. For the beam wave direction the results given by Ogawa for the captive vessel are compared in Fig. 5 with the experimental values of the wave drift force on a free floating vessel having a smaller block coefficient of 0.60. In this Figure the drift force Y_d is given in a non-dimensional way by dividing the force by $\frac{1}{2} \rho_w g \cdot \zeta_a^2 \cdot L$ and taking the square root. (ζ_a = amplitude of incident wave).

This non-dimensional drift force coefficient

$$R = \sqrt{\frac{Y_d}{\frac{1}{2} \rho_w g \zeta_a^2 \cdot L}}$$

is plotted to a base of non-dimensional wave length $k.T$, in which:

$$k = \frac{2\pi}{\lambda} = \text{wave number}$$

λ = wave length

All results apply to deep water. As illustrated in Fig. 5 the drift force on the free floating vessel corresponds quite well with the force on an infinitely long flat vertical plate (see (10)) with a draft equal to the vessel's draft and over a length that equals the length of the vessel between perpendiculars.

The drift force on the restrained vessel in deep water can be compared quite well with the results given of theoretical calculations conducted by Mei and Black (11) for a rectangular captive cylinder extrapolated to the same beam over draft ratio ($B/T = 2.5$) as the vessel has and to deep water. This indicates that the influence of diffraction due to the restricted length/beam ratio of the vessel probably may be neglected.

The considerable larger drift force on the captive rectangular cylinder relative to the flat plate demonstrates the important effect of the beam or the bottom of the vessel and consequently of the relative vertical motion between the object and the wave motion. The rather good agreement between the free floating vessel and the vertical plate seems to indicate that the drift force contributed by the relative heave motion of the vessel is small in this particular case. The influence of the water depth-draft ratio is indicated in Figure 6, where some results are given of measurements carried out at the N.S.M.B. on a series 60 model, block 0.80, beam/draft ratio 2.5, length/beam ratio 7. From this figure it will be clear that in deep water the drift force in beam waves on the block 0.80 vessel corresponds also reasonably well with the

force on a vertical plate. However, at the very reduced water depth of 1.1 times the draft of the vessel, very high drift forces are measured for wave frequencies near the natural frequency of the roll motion. This may be explained by the much higher roll damping (the damping coefficient is approx. 3 times larger) at this water depth, which means that the roll generated waves are higher and consequently the steady drift force too.

For an approximation of the lateral drift force in deep water in regular waves the following expression may be used:

$$Y_d = \frac{1}{2} \rho_w g \zeta_a^2 \cdot R^2 \cdot L \cdot \sin^2 \alpha$$

in which

ζ_a = amplitude of incident wave

R = drift force coefficient for a vertical plate with draft T . R is a function of the dimensionless wave length kT

α = wave direction

For the determination of the mean drift force or the resistance in head waves reference is made to the method described by Gerritsma and Beukelman (12). This method is based on the determination of the radiated energy by calculating the amplitude of the waves generated by the relative vertical motion between ship and waves. In case of a rather flat bow also the influence of the relative surge motion has to be taken into account as is shown in (13).

The wave drift force on other structures

Up to now only the drift force on ship shaped bodies have been dealt with.

For the drift force on semi submersible

type structures no data are available on this moment. Probably the best way to estimate the drift force is to split up the construction in elements consisting of circular or rectangular cylinders and to estimate the drift force on each element separately. In (14) data are given for the drift force on a restrained vertical cylinder.

Wave drift force in irregular waves

When the drift force on a structure is known as a function of the wave frequency either from calculations or model tests, the lateral mean drift force in irregular waves, described by a particular wave spectrum, can be determined from:

$$\bar{Y}_d = \rho_w g \cdot L \cdot \int_0^\infty S_\zeta(\omega) \cdot \frac{Y_d(\omega)}{\frac{1}{2} \rho g \zeta_a^2} d\omega$$

As an empirical approximation for ship shaped bodies the following expression may be used for an irregular sea described by a narrow spectrum:

$$\bar{Y}_d = \frac{1}{16} \rho_w g \zeta_w^2 \frac{1}{3} \cdot R^2(\bar{\omega}) \cdot L \sin^2 \alpha$$

in which

$\zeta_w \frac{1}{3}$ = significant wave height (crest-trough)

$R(\bar{\omega})$ = drift force coefficient for flat plate for $\omega = \bar{\omega}$ = mean wave frequency

Design of the anchor system

If the steady force on a structure in a particular sea state is known and the lay-out of the anchor system has been selected the minimum thickness of the anchor lines is determined by the mean external force as will be illustrated below.

Suppose the anchor system has to satisfy the following criteria

1. The maximum allowable excursion $x_{\max}$ of the structure from its initial unloaded equilibrium position is given as a certain percentage of the water depth.
2. The maximum allowable tension in the anchor lines may not exceed a certain percentage of the breaking strength.

The thickness of the anchor chains is proportional to the weight per unit length (specific weight). The minimum weight required will be attained when the pretension in the anchor lines is such that at an excursion which equals the maximum allowable excursion also the maximum allowable load in the heaviest loaded anchor line is attained.

From the non-dimensional catenary characteristics the tension in the line T_0 divided by $w \cdot h_a$ and the angle θ between the line and the horizontal plane, both measured at the attachment point of the line to the structure, can be determined as a function of the excursion x/h_a of the structure.

w = submerged weight of anchor line per unit length

h_a = height of attachment point above bottom

Since the breaking strength of a particular type of anchor line is proportional to the specific weight w , the excursion of the structure can be determined at which the tension $T_0/w \cdot h_a$ equals the maximum allowable tension. Then the required pretension angle θ can be read off at an excursion which is $x_{\max}$ smaller. The usual non-linear relationship between the excursion of the structure and the horizontal load $F_H/w \cdot h_a$ required for that excursion can be calculated for the

selected pretension.

Subtracting the maximum expected fluctuating motion from the maximum allowable excursion $x_{\max}$ gives the excursion of the structure which can be allowed as a result of the mean force $\bar{F}$ due to wind, waves and current. The total horizontal force $F_H' = F_H/w.h_a$ at this excursion can be read of.

Then the minimum required submerged weight of the anchor lines has to satisfy

$$w = \frac{\bar{F}}{F_H' \cdot h_a}$$

The above described procedure will be illustrated by means of an example.

Example

Question Determine the minimum weight of the anchor lines of an anchoring system for a turret mooring of a drilling vessel having the following main dimensions:

length = 150 m
 beam = 26 m
 draft = 8 m
 wind exposed transverse area = 750 m²
 displacement = 17500 metric tons
 water depth = 70 m

The design has to be based on a sea state described by a Pierson-Moskowitz spectrum with a significant wave height of 4.5 m and a mean period of 8 sec. The maximum wind speed is 40 knots, the current speed 2 knots. The maximum allowable excursion is 9% of the water depth. The forces in the lines may not exceed 50% of the breaking strength. Although the vessel is equipped with bow and stern thrusters to control the heading, since the turret is located amidships, the anchor system has to be designed for the combined action

of beam wave, wind and current. The system consists of 8 anchor legs equally distributed over the circumference of the turret. From model test data on similar ships the maximum oscillatory excursion of the vessel is estimated to be approximately 4.35 m being 7% of the height h_a of the attachment points of the anchor chains above the bottom.

Solution

The mean force on the vessel determined according the data given in this paper are as follows:

due to a 40 knots beam wind	17 ton
due to a 2 knots beam current	49 ton
due to 4.5 m significant height waves	48 ton
Total	114 ton

From the characteristics of the catenary it can be determined that the anchor legs have to consist of approx. 500 m (8 x h) anchor chain (U-3 quality) or 1050 m (17 x h) steelwire (h = water depth-draft = 62 m) in order to be sure that the tangent of the line at the anchor coincides with the bottom at a tension which equals half the breaking strength.

The pretension angle θ , required to obtain a tension which equals half the breaking strength at the maximum allowable excursion of 9% of the height h_a , amounts to the following values

for 500 m U-3 quality stud link chain:
 $\theta_{\text{pret.}} = 26.2$ degrees
 for 1050 m steelwire: $\theta_{\text{pret.}} = 10.7$ degrees.

The non-dimensional relationships between the excursion of the vessel and the horizontal force required have been calculated for both types of anchor lines. For U-3 stud link chain the result is

shown in Fig. 6. At an excursion of 9% of the height h_a , the tension in the heaviest loaded anchor line attains half the breaking strength T_{obr}

for U-3 quality chain $T_{obr} \approx 4000 \times w$
for steelwire $T_{obr} \approx 17500 \times w$

T_{obr} in kg
 w in kg'm⁻¹

Subtracting the maximum expected oscillatory excursion from the maximum allowable excursion leads to an excursion of 2% of the height h_a , that may be allowed as a result of the mean force of 108 ton on the vessel. The dimensionless horizontal load on the system corresponding to this 2% excursion amounts to:

for U-3 chain 8.7

for steelwire 59.1

The resulting minimum required submerged weight of the anchor lines is given in Table VI. The corresponding approximate diameters of the lines are also mentioned in Table VI.

The method described here has to be adapted for each special case. However, the example illustrates clearly the important role which the mean force may play for the determination of the anchor system.

Nomenclature

a_n, b_n Fourier coefficients
 h depth of water
 h_a height of attachment point of anchor line above bottom
 k wave number = $2\pi/\lambda$
 p pressure
 p_o atmospheric pressure
 t time
 w submerged weight of anchor line
 x excursion

x_{max} maximum allowable excursion of structure
 x, y, z right handed system of coordinates
 z vertical coordinate, upward positive
 A_L lateral area above water surface
 A_{LS} submerged lateral area
 A_T transverse area above water surface
 A_{TS} submerged transverse area
 B breadth of ship
 C_{nw} yaw wind moment coefficient
 C_{nc} yaw current moment coefficient
 C_{xw} longitudinal wind force coefficient
 C_{xc} longitudinal current force coefficient
 C_{yw} transverse wind force coefficient
 C_{yc} transverse current force coefficient
 $\bar{F}$ total mean load on anchored structure
 F_H horizontal load on anchor system
 N_C yaw current moment
 N_W yaw wind moment
 R non-dimensional drift force coefficient
 T draft of ship
 T_O tension in anchor line
 V velocity of water motion
 V_w wind velocity
 V_c current velocity
 X_C longitudinal current force
 X_w longitudinal wind force
 Y_C transverse current force
 Y_d transverse drift force
 $\bar{Y}_d$ mean transverse drift force in irregular waves

Y_w	mean transverse wind force
α	angle of incidence
ζ_a	amplitude of incident wave
ζ_{ar}	amplitude of reflected and scattered wave
$\zeta_w^{1/3}$	significant wave height
θ	angle between anchor line and horizontal plane at the attachment point to the structure
θ_{pret}	pretension angle = angle θ for T_o is equal to pretension
ϕ	velocity potential
ρ_a	density of air
ρ_w	density of water
ω	wave frequency
$\bar{\omega}$	mean wave frequency of an irregular sea

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Publication no. 375 of the N.S.M.B.

TABLE 1 - DATA OF TANKERS

Ship No.	Type	Length	Condition	Data taken from ref.
1	bridge amidsh.	-	loaded	2
2	" "	-	ballast	2
3	bridge aft	-	loaded	2
4	" "	-	ballast	2
5	bridge amidsh.	225 m.	loaded	3
6	" "	"	ballast	3
7	bridge aft	"	loaded	3
8	" "	"	ballast	3
9	" "	172 m.	loaded	4
10	" "	150 m.	loaded	5
11	" "	"	ballast	5

TABLE 2 - COEFFICIENTS FOR THE LONGITUDINAL FORCE ON TANKERS DUE TO WIND

Ship No.	a_0	a_1	a_2	a_3	a_4	a_5
1	- 0.131	0.738	- 0.058	0.059	0.108	- 0.001
2	- 0.079	0.615	- 0.104	0.085	0.076	0.025
3	- 0.028	0.799	- 0.077	- 0.054	0.018	- 0.018
4	0.014	0.732	- 0.055	- 0.017	- 0.018	- 0.058
5	- 0.074	1.050	0.017	- 0.062	0.080	- 0.110
6	- 0.055	0.748	0.018	- 0.012	0.015	- 0.151
7	- 0.038	0.830	0.031	0.012	0.021	- 0.072
8	- 0.039	0.646	0.034	0.024	- 0.031	- 0.090
9	- 0.042	0.487	- 0.072	0.109	0.075	- 0.047
10	- 0.075	0.711	- 0.082	0.043	0.064	- 0.038
11	- 0.051	0.577	- 0.058	0.051	0.062	0.006

TABLE 3 - COEFFICIENTS FOR THE TRANSVERSE FORCE ON TANKERS DUE TO WIND

Ship No.	b_1	b_2	b_3	b_4	b_5
1	0.786	0.039	0.003	0.034	- 0.019
2	0.880	0.004	0.003	- 0.004	- 0.003
3	0.697	0.036	0.018	0.028	- 0.023
4	0.785	0.014	0.014	0.015	- 0.020
5	0.707	- 0.013	0.028	0.007	- 0.044
6	0.731	- 0.014	0.016	0.001	- 0.025
7	0.718	0.032	0.010	- 0.001	- 0.040
8	0.735	0.003	0.004	- 0.005	- 0.017
9	0.764	0.037	0.052	0.016	- 0.003
10	0.819	0.051	0.023	0.032	- 0.032
11	0.879	0.026	0.014	0.031	- 0.029

TABLE 4 - COEFFICIENTS FOR THE YAW MOMENT ON TANKERS DUE TO WIND

Ship No.	b_1	b_2	b_3	b_4	b_5
1	- 0.0451	- 0.0617	- 0.0110	- 0.0110	- 0.0010
2	- 0.0338	- 0.0800	- 0.0080	- 0.0096	- 0.0013
3	- 0.0765	- 0.0571	- 0.0166	- 0.0146	0.0021
4	- 0.0524	- 0.0738	- 0.0175	- 0.0089	- 0.0021
5	- 0.0216	- 0.0531	- 0.0063	- 0.0073	0.0024
6	- 0.0059	- 0.0730	- 0.0035	- 0.0017	- 0.0013
7	- 0.0526	- 0.0596	- 0.0111	- 0.0113	0.0099
8	- 0.0335	- 0.0722	- 0.0090	- 0.0047	0.0067
9	- 0.1025	- 0.0721	- 0.0345	- 0.0127	- 0.0022
10	- 0.0881	- 0.0681	- 0.0202	- 0.0145	0.0039
11	- 0.0644	- 0.0726	- 0.0244	- 0.0076	0.0024

TABLE 5 - COEFFICIENTS FOR THE TRANSVERSE FORCE AND YAW MOMENT ON TANKERS DUE TO CURRENT FORCE FOR THE LOADED CONDITION IN DEEP WATER

	transverse force	yaw moment
b_1	0.908	- 0.0252
b_2	0.000	- 0.0904
b_3	- 0.115	0.0032
b_4	0.000	0.0109
b_5	- 0.033	0.0011

TABLE 6 - THE MINIMUM REQUIRED SUBMERGED WEIGHT AND THE APPROXIMATE DIAMETERS OF THE ANCHOR LINES

	submerged weight per meter length	diameter
U-3 stud link chain	211 kg	107 mm $\approx$ 4 1/4"
steelwire	31.2 kg	92 mm $\approx$ 3 5/8"

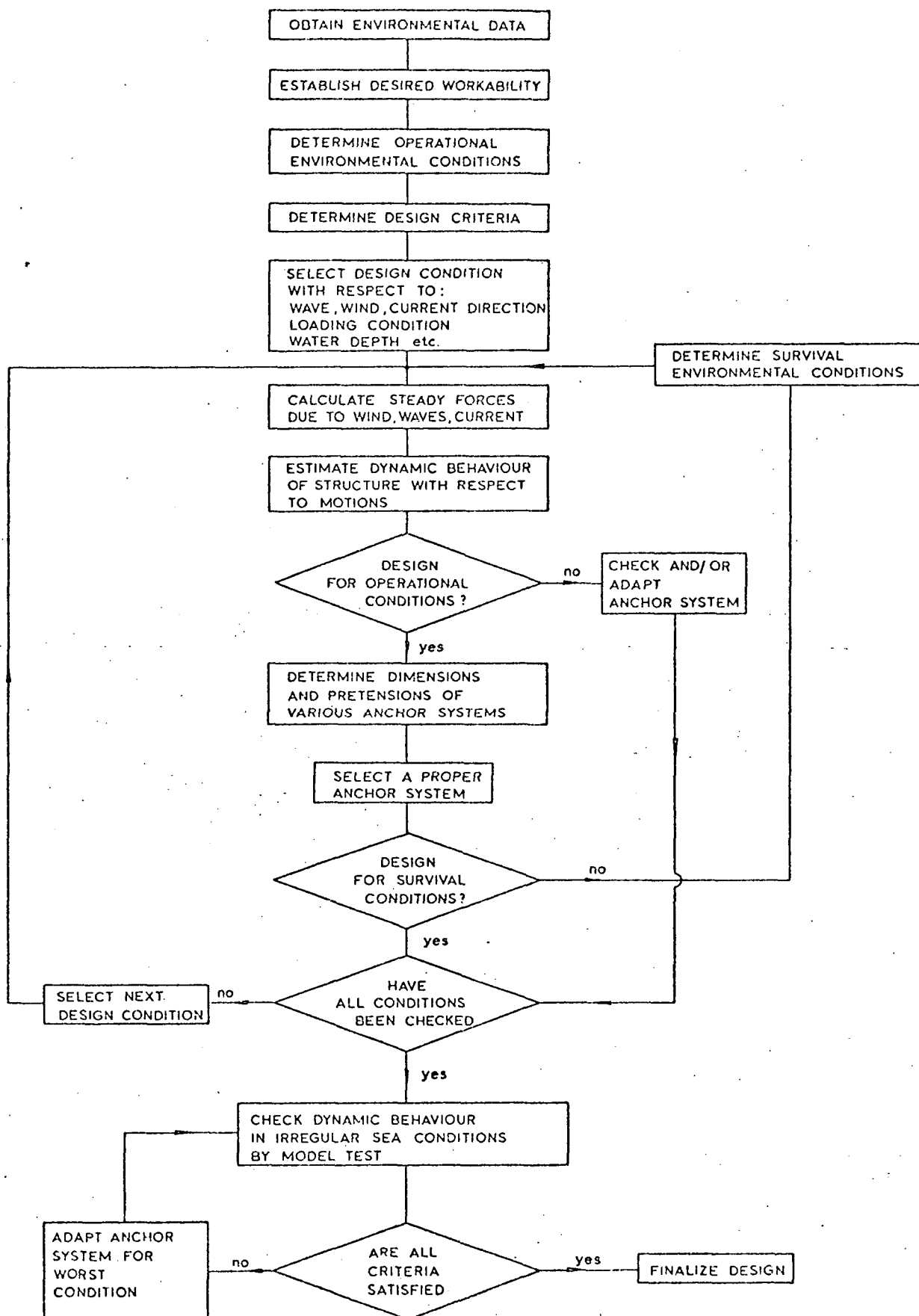


Fig. 1 - The design of an anchor system.

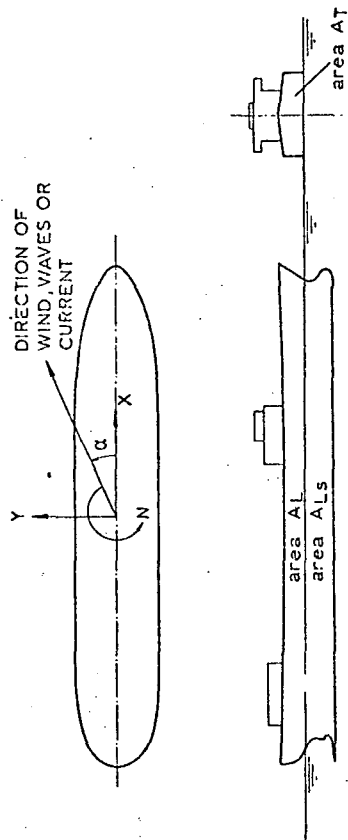


Fig. 2 - Description of forces and moments.

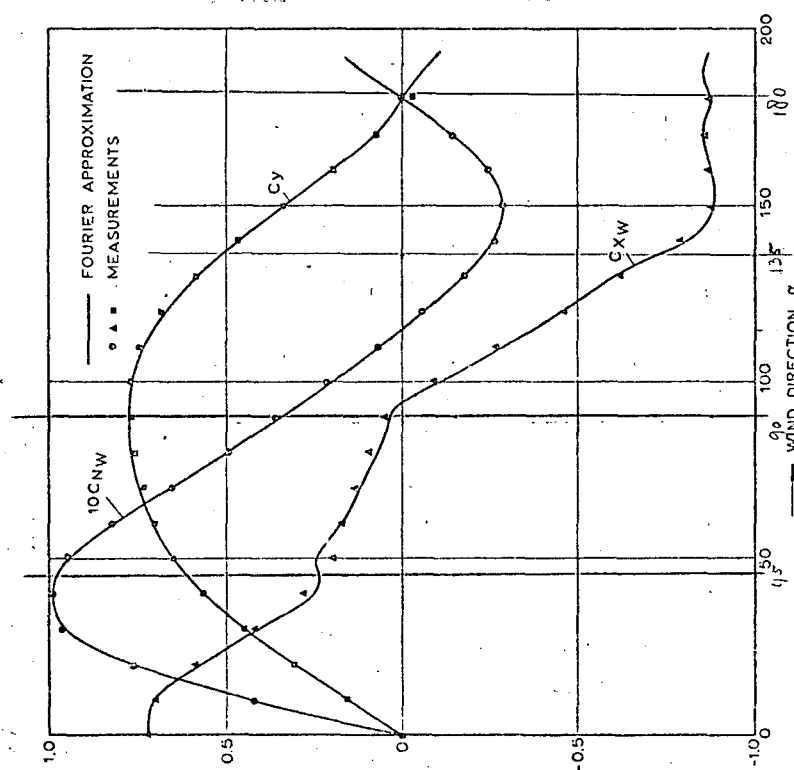


Fig. 3 - Comparison of the Fourier approximation with measurements of the wind load on the tanker.

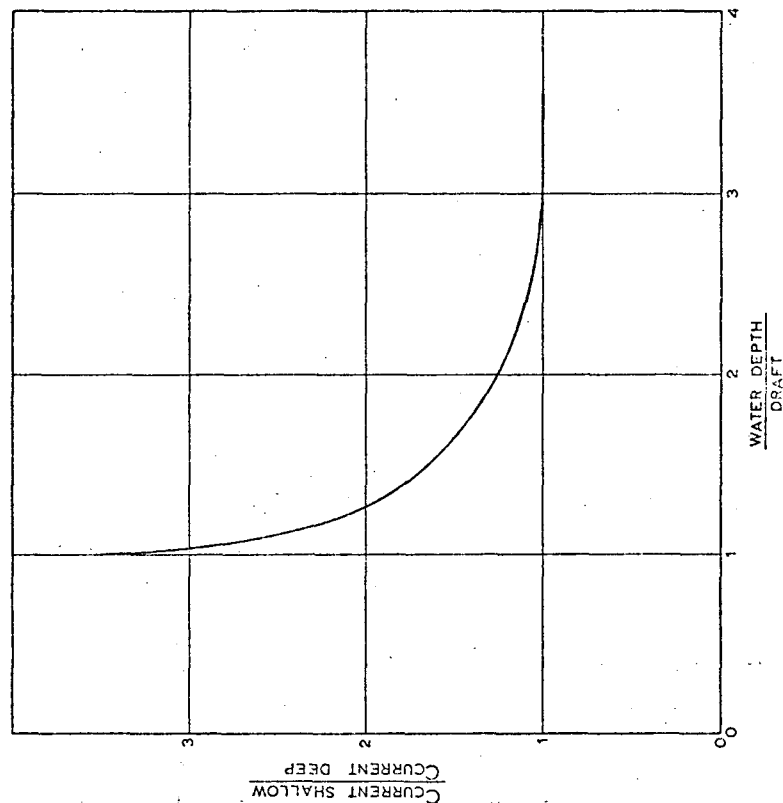


Fig. 4 - The influence of the water depth on the current load on a tanker.

DRIFT FORCE COEFFICIENT $R = \sqrt{\frac{\gamma_d}{\frac{1}{2} \rho_w g \zeta_a^2 L}}$
DEEP WATER

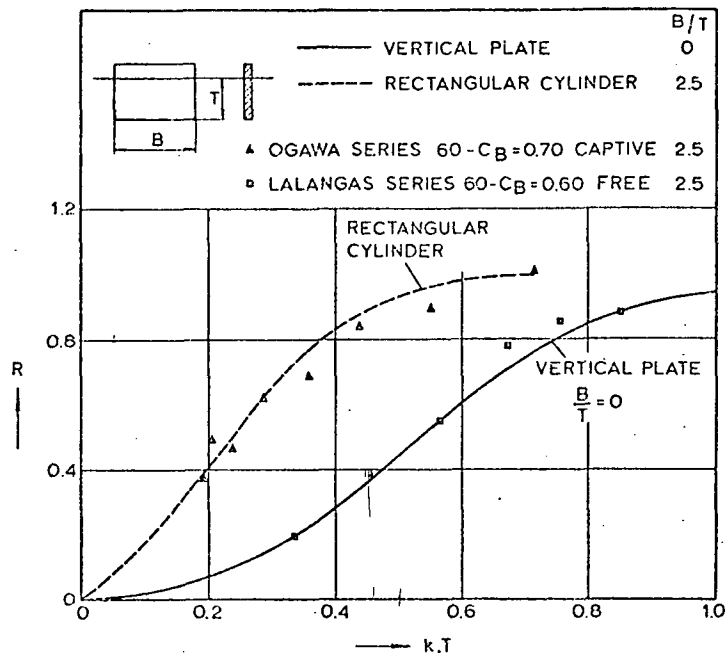


Fig. 5 - Drift force coefficient in beam waves.

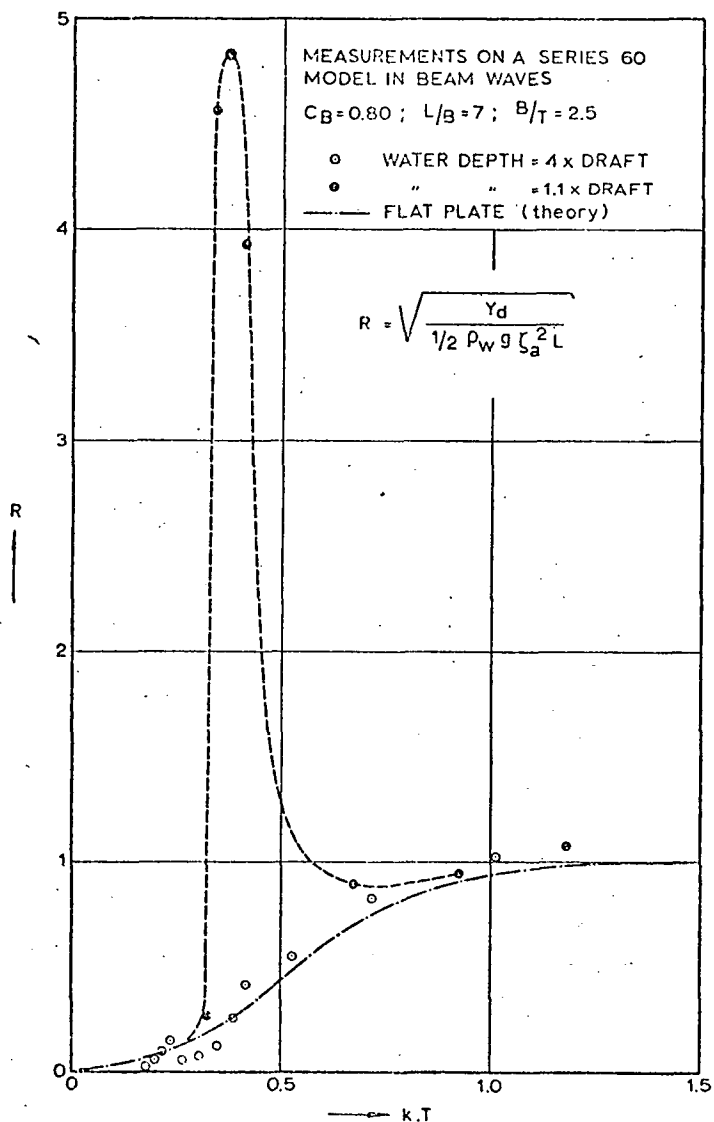


Fig. 6 - Influence of water depth on the drift force coefficient.

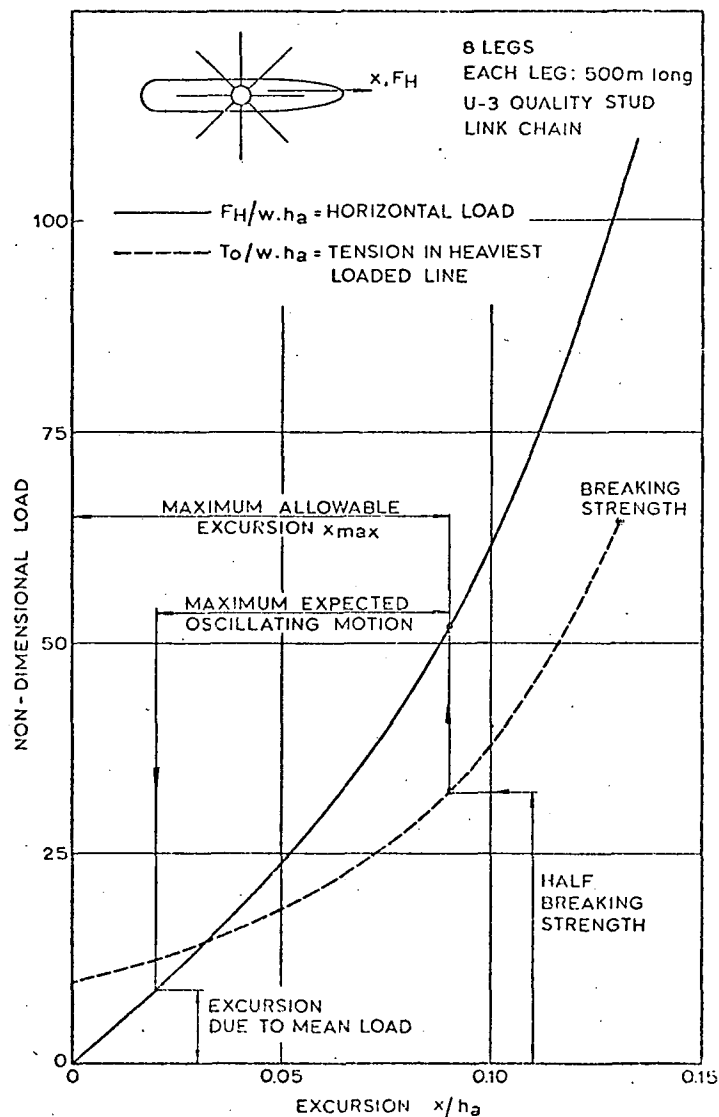


Fig. 7 - Characteristics of the eight-leg anchor system.

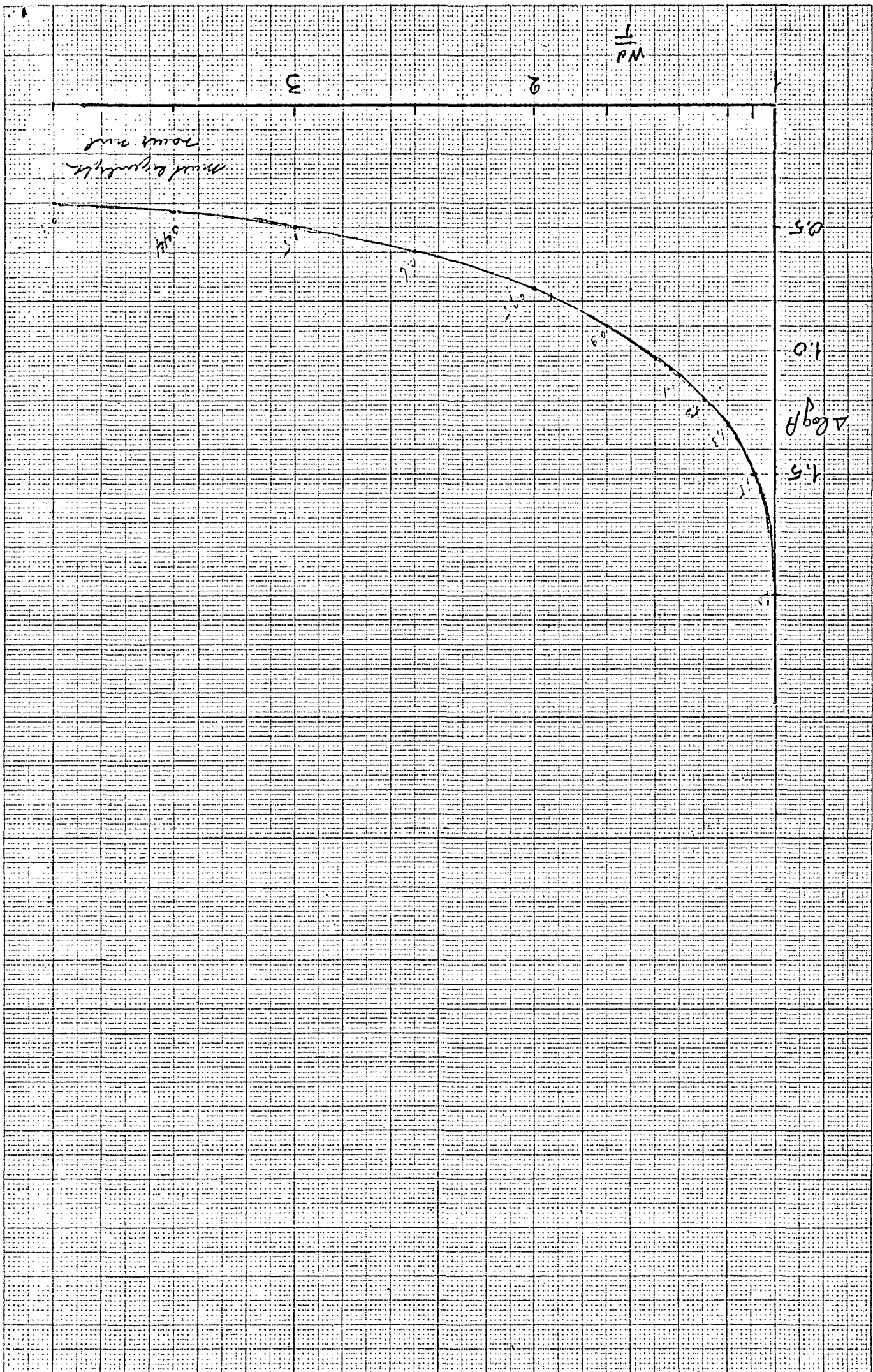


fig. 8

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The Mean Wave, Wind and Current Forces on Offshore Structures and Their Role in the Design of Mooring Systems

By

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Abstract

The mean forces induced by waves, wind and current on tankers, barges and other structures, are very important for the selection of an appropriate passive or active positioning system. In the present paper tools are given for the determination of these forces, especially for tanker shaped bodies, taking into account the main influence of water depth. The way in which the estimation of the mean forces has to be used for a preliminary design of a mooring system will be dealt with.

Introduction

The problem of designing an adequate mooring or anchoring system for floating offshore structures is rather complicated not at least due to the non-linear nature

of the problem. It is a problem of dynamics and the moored structure may be regarded as a mass-spring system. The total load, exerted by the environment, consists of the oscillating wave forces and of forces, which vary at a frequency much lower than the wave frequency: the wind, current and wave drift forces. When designing an anchor system, three aspects of these very slowly varying forces are of importance.

1. As will be illustrated at the end of this paper the almost steady wind, current and wave drift forces are most significant for the final determination of the diameter of the anchor lines. This is due to the fact that the main purpose of an anchoring system is to hold the structure on an average position. Basically the force required for this station keeping equals the steady or the slowly

oscillating wave, wind and current forces on the structure.

2. The almost steady wind, current and wave drift forces cause a shift of the neutral position of the moored object, around which the oscillation due to waves occurs. Since the relationship between the horizontal displacement of the anchored structure and the restoring force of the anchor system is almost always non-linear, the shift of the neutral position causes a change in the spring rate, and consequently in the dynamic response. However, generally the natural frequency of the horizontal motion of the anchored structure is considerably smaller than the wave frequency, also after an increase of the spring constant, due to the mean shift of the structure, but this has to be checked for each special design. Therefore it is very important to study this effect of the wind, current and wave drift forces in an early stage of the design, to be sure that resonance at the wave frequency will be avoided.
3. Since the wind, current and wave drift forces are slowly fluctuating forces, the risk exists that resonance occurs at the very low frequencies, at which these forces oscillate. Low frequency oscillations in the horizontal plane have been observed in model experiments as well as at full scale. The nature of this phenomenon is not yet fully understood, but in general it can be attributed to the slowly varying wave drift force as a result of the varying wave height in irregular seas. Attention has to be paid to the dynamic forces due to wind gusts,

while the variation in the current velocity occurs at a much too low frequency to be of importance.

The first two aspects discussed here, have to be taken into account during the preliminary design, and will be of significant importance for this design. For this aspect the wind, current and wave drift may be regarded as steady phenomena, inducing only constant forces. For the investigation of the dynamic character of especially the wave drift force and its effect on the dynamic response of the anchored object, model tests are still indispensable.

Figure 1 shows a flow diagram, depicting a possible design process for an anchor system. In general, the process will have to be repeated for a number of different conditions. For a drill rig, for instance the anchor system must be strong enough to withstand the extreme loads in the survival condition, while for the maximum operational wave condition the motion of the rig may not exceed a certain value. In a certain condition, it may be necessary to consider various combinations of wave, current and wind directions.

In this paper we will deal with the determination of the steady forces, mainly on tanker hulls. The nature of the wind, current and wave drift forces will be briefly discussed and data will be given for an estimate of these forces. The problem of the choice of the design environmental conditions will be left out of consideration. An example how to use these data for the design of an anchor system will be given.

Description of steady forces and moments

Wind forces and current forces are frequently presented as being composed of a drag force, in the direction of the flow, and a lift force, perpendicular to the flow direction. However, when describing the dynamic behaviour of a floating object, the equations of motion are usually related to a structure bound system of coordinates, with its origin in the midship section of the structure. Therefore it is convenient to describe the steady forces as a longitudinal and a transverse component applying in the midship section and a moment around the vertical axis. Figure 2 shows a sketch in which the sign of the forces and moment, and the direction of wind, waves or current are defined.

The force due to wind

Like all environmental phenomena, wind has a stochastic nature which greatly depends on time and location. It is usually characterized by fairly large fluctuations in velocity and direction. It is common meteorological practice to give the wind velocity in terms of the average over a certain interval of time, varying from 1 to 60 minutes or more. The variation in mean velocity is very slow compared with the wave period. The fluctuations around the mean value will impose dynamic forces on the structure, but in general these aerodynamic forces may be neglected in comparison with the hydrodynamic forces, when considering the dynamic behaviour of the floating body. Therefore, we will consider the wind velocity as a steady value, both in magni-

tude and direction, resulting in constant forces and a constant moment acting on the structure.

The role which wind plays in the determination of the environmental conditions of floating structures is twofold:

- On the part of the structure exposed to the air, wind forces are exerted due to the stream of air around the various parts. For the determination of these forces information is required about the local winds only.
- The forces exerted by the wind on the water surface cause disturbances of the still water level, thus generating waves and current, which induce forces on the submerged part of the structure. To determine this effect of the wind, information is required about the wind and storm conditions in a much larger area.

The wave and current generating character of the wind will be left out of consideration. The effects of waves and currents will be dealt with separately.

Local winds are generally defined in terms of the average velocity and average direction related to a certain height above the still water level. The standard height above the water surface for which generally the wind velocity data are given amounts to 10 metres or 30 feet. A number of empirical and theoretical formulas is available in literature to determine the wind velocity at other heights. An adequate vertical distribution of the wind speed is represented by (see ref. (1)):

$$\frac{V(z)}{V(10)} = \left(\frac{z}{10}\right)^{1/7}$$

in which

$V(Z)$ = wind speed at height Z above the water surface

$V(10)$ = wind speed at 10 metres height above the water surface

The total force and moment experienced by an object exposed to wind, is partly of viscous origin (pressure drag) and partly due to potential effects (lift force and moment). For blunt bodies, the wind force may be regarded as independent of the Reynolds number and proportional to the square of the wind velocity.

Wind load data for tankers

The forces and moment exerted by wind on tankers can be calculated from:

$$\begin{aligned} X_w &= \frac{1}{2} \rho_a V_w^2 C_{xw}(\alpha) \cdot A_T \\ Y_w &= \frac{1}{2} \rho_a V_w^2 C_{yw}(\alpha) \cdot A_L \\ N_w &= \frac{1}{2} \rho_a V_w^2 C_{nw}(\alpha) \cdot A_L \cdot L \end{aligned}$$

in which

X_w = steady longitudinal wind force

Y_w = steady transverse wind force

N_w = steady yaw wind moment

ρ_a = density of air

V_w = wind velocity

α = wind direction

A_T = exposed transverse area

A_L = exposed lateral area

L = length of the ship

C_{xw} , C_{yw} and C_{nw} are coefficients, depending on the angle of incidence of the wind.

In literature the results of wind tunnel experiments are given for various types of vessels. From several papers the wind data on tanker hulls were collected, and the force and moment coefficients C_{xw} , C_{yw} and C_{nw} were expanded in Fourier series as a function of the angle of incidence:

$$C_{xw} = a_0 + \sum_{n=1}^{\infty} a_n \cos n\alpha$$

$$C_{yw} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

$$C_{nw} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

From the harmonic analysis it was found that a fifth order representation of the wind data is sufficiently accurate for preliminary design purposes.

In Table II through IV the Fourier coefficients are given for the longitudinal and transverse force, and the yawing moment, for a number of tankers. Particulars of the tankers are listed in Table I, where also the reference numbers are given of the publications, from which the data were taken. Figure 3 shows, as an example, the measured wind forces and moment together with the Fourier approximation, for one of the tankers.

The wind load on other structures

The wind forces and moments on other types of structures, as for instance floating semi-submersible platforms, can be approximated by dividing the structure in a number of components, all with a more or less elementary geometry, and estimating the wind force on each element. For a lot of simple geometrical forms such as spheres, flat plates and cylinders of various cross sectional shapes, the drag and lift coefficients are given in literature. Reference is made of the publications of Hoerner (6) and Delany and Sorensen (7). The total wind load on the structure is then found by adding the contributions of all the component parts.

The force due to current

There are several independent phenomena responsible for the occurrence of current: the ocean circulation system, resulting in a steady current, the cyclical change in lunar and solar gravity, causing tidal currents, wind and differences in density. The steady velocity at the water surface due to wind amounts to about 3 percent of the wind velocity (at 30 ft height). Tidal currents are of main importance in areas of restricted water depth and can attain values up to 10 knots. However, these extreme velocities are rare. A 2 or 3 knots tidal current speed is common. The prediction of the magnitude of tidal currents is a special science. Some predictors take not less than 60 parameters into account in their prediction procedure.

Although for floating structures the surface currents will be the governing ones, the vertical current distribution may also be of importance, especially for the case of restricted water depth. For the design of an anchor system of a floating structure the designer is especially interested in the probability that a particular extreme current velocity will be exceeded during a certain period of time. Observations obtained from current speed measurements are indispensable for that purpose. It may be useful to split up the total measured current in two or more components, for instance in a tidal and a non-tidal component, since the direction of the various components will be different, in general. The variation in velocity and direction of the current is very slow, and current may therefore be considered

as a steady phenomenon.

The forces and moment exerted by current on a floating object is composed of the following parts:

- A viscous part, due to friction between the structure and the fluid, and due to pressure drag. For blunt bodies the frictional force may be neglected, since it is small compared to the viscous pressure drag.
- A potential part, with a component due to a circulation around the object, and an other component due to the free water surface (wave resistance). In most cases, the latter component is small in comparison with the first.

Current load data for tankers

The forces and moment exerted by current on a tanker hull can be described by:

$$\begin{aligned} X_c &= \frac{1}{2} \rho_w V_c^2 C_{xc}(\alpha) \cdot A_{TS} \\ Y_c &= \frac{1}{2} \rho_w V_c^2 C_{yc}(\alpha) A_{LS} \\ N_c &= \frac{1}{2} \rho_w V_c^2 C_{nc}(\alpha) A_{LS} \cdot L \end{aligned}$$

in which

X_c = steady longitudinal current force
 Y_c = steady transverse current force
 N_c = steady yaw current moment
 ρ_w = density of water
 V_c = current velocity
 α = angle of incidence
 A_{TS} = submerged transverse area
 A_{LS} = submerged lateral area = $L \times T$
 L = length of the ship
 T = draft of the ship
 C_{xc} , C_{yc} and C_{nc} are coefficients, depending on the current direction.

At the N.S.M.B., the current loads have been measured on several tanker models of different size. From the results the

coefficients C_{xc} , C_{yc} and C_{nc} were calculated. For flow in the longitudinal direction a tanker hull is a rather slender body, and consequently consists the longitudinal force mainly of frictional resistance. The total longitudinal force was very small for the relatively low current speed, and could therefore not be measured very accurately. Moreover, extrapolation to full scale dimensions is difficult, since the longitudinal force is affected by scale effect. For mooring problems, the longitudinal current force will hardly be of importance. An estimate of its magnitude can be made by calculating the flat plate frictional resistance:

$$X_c = \left(\frac{0.075}{(\log R_n - 2)^2} \right) \cdot \frac{1}{2} \rho_w S V_c^2 \cos \alpha |\cos \alpha|$$

in which

$$R_n = \text{the Reynolds number} = \frac{V_c \cos \alpha}{\nu L}$$

ν = kinematic viscosity of water

S = the wetted surface

Extrapolation of the transverse force and yaw moment to prototype values is no problem. For flow in the transverse direction the tanker is a blunt body, and since the bilge radius is small, flow separation occurs in the model in the same way as in the prototype. Therefore, the transverse force coefficient and the yaw moment coefficient are independent of the Reynolds number.

The coefficients for the transverse force and the yaw moment were expanded in a Fourier series, as was done for the wind load coefficients:

$$C_{yc} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

$$C_{nc} = \sum_{n=1}^{\infty} b_n \sin n\alpha$$

The average value of the Fourier coefficients for the fifth order Fourier series, are given in Table V. These results apply to deep water. For shallow water, the current force and moment coefficients have to be multiplied by a coefficient, which is given in Figure 4 on a base of the water depth-draft ratio. In the data, given in Table V, the influence of the free water surface is included. This influence, however, depends on the water depth and on the Froude number, and consequently changes if the current velocity or the tanker dimensions change. For the condition to which these data apply, deep water and a prototype current speed in the order of 3 knots, the effect of the free water surface is very small. For a case of a small underkeel clearance and a current direction of 90 degrees damming up of the water at the weather-side and a lowering of the water level at the lee side of the ship occurs.

The current load on other structures

The current load on other types of floating structures can be estimated in the same way as was described for the wind load in a previous section.

Wave drift forces

A structure floating in waves experiences forces and moments which can be determined if the velocity potential ϕ of the water motion around the structure is known. By integrating the component of the pressure in a particular direction over the hull of the structure, the force component in this direction can be calculated. The pressure can be obtained

from the Bernoulli equation for non-steady flow:

$$p = p_0 + \rho_w \frac{\partial \phi}{\partial t} - \rho_w g z + \frac{1}{2} \rho_w V^2$$

in which

p = pressure

p_0 = atmospheric pressure

ϕ = velocity potential

g = acceleration of gravity

V = velocity of water motion

$$V^2 = \left(\frac{\partial \phi}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial y} \right)^2 + \left(\frac{\partial \phi}{\partial z} \right)^2$$

In the theory of periodic ship's motion in waves the velocity term $\frac{1}{2} \rho_w V^2$ is neglected, since it has only a second order influence on the oscillatory behaviour. However, it is this term which is responsible for the steady drift force. Representing the velocity potential by a periodic function proportional to the amplitude ζ_a of the incident wave

$$\phi = \phi'_a \cdot \zeta_a \cdot \sin \omega t$$

in which

ω = frequency of waves

it follows that

$$\frac{1}{2} \rho_w V^2 = \frac{1}{2} \rho_w \cdot \zeta_a^2 \cdot \sin^2 \omega t \left\{ \left(\frac{\partial \phi'_a}{\partial x} \right)^2 + \left(\frac{\partial \phi'_a}{\partial y} \right)^2 + \left(\frac{\partial \phi'_a}{\partial z} \right)^2 \right\}$$

By taking the average value of the pressure during one period ($T = \frac{2\pi}{\omega}$) it will be clear that all periodic terms vanish and only the velocity term $\frac{1}{2} \rho_w V^2$ gives a contribution. Consequently the steady drift force on a structure in waves is proportional to the square of the height of the incident wave. Maruo (8) shows that the lateral drift force Y'_d per unit length on an infinitely long cylinder

(2-dimensional case; no diffraction) in beam seas satisfies:

$$Y'_d = \frac{1}{2} \rho_w g \zeta_{ar}^2$$

in which

ζ_{ar} = amplitude of wave reflected and scattered by the body

He also indicates that the amplitude of this wave is proportional to the relative motion between the oscillating cylinder and the wave.

Wave drift load data for tankers

Ogawa (9) applied Maruo's theory on a captive series 60 model (block coefficient 0.70) in beam and bow quatering waves and he shows that

$$Y'_d = \frac{1}{2} \rho_w g \zeta_{ar}^2 \cdot \sin^2 \alpha$$

in which

α = direction of waves relative to ship

He calculated the amplitude of the reflected and scattered wave using the strip theory for two wave directions (90° and 120°) relative to the captive vessel and found a reasonable agreement with the measured results. For the beam wave direction the results given by Ogawa for the captive vessel are compared in Fig. 5 with the experimental values of the wave drift force on a free floating vessel having a smaller block coefficient of 0.60. In this Figure the drift force Y_d is given in a non-dimensional way by dividing the force by $\frac{1}{2} \rho_w g \cdot \zeta_a^2 \cdot L$ and taking the square root. (ζ_a = amplitude of incident wave).

This non-dimensional drift force coefficient

$$R = \sqrt{\frac{Y_d}{\frac{1}{2} \rho_w g \zeta_a^2 \cdot L}}$$

is plotted to a base of non-dimensional wave length $k.T$, in which:

$k = \frac{2\pi}{\lambda}$ = wave number

λ = wave length

All results apply to deep water. As illustrated in Fig. 5 the drift force on the free floating vessel corresponds quite well with the force on an infinitely long flat vertical plate (see (10)) with a draft equal to the vessel's draft and over a length that equals the length of the vessel between perpendiculars. The drift force on the restrained vessel in deep water can be compared quite well with the results given of theoretical calculations conducted by Mei and Black (11) for a rectangular captive cylinder extrapolated to the same beam over draft ratio ($B/T = 2.5$) as the vessel has and to deep water. This indicates that the influence of diffraction due to the restricted length/beam ratio of the vessel probably may be neglected.

The considerable larger drift force on the captive rectangular cylinder relative to the flat plate demonstrates the important effect of the beam or the bottom of the vessel and consequently of the relative vertical motion between the object and the wave motion. The rather good agreement between the free floating vessel and the vertical plate seems to indicate that the drift force contributed by the relative heave motion of the vessel is small in this particular case. The influence of the water depth-draft ratio is indicated in Figure 6, where some results are given of measurements carried out at the N.S.M.B. on a series 60 model, block 0.80, beam/draft ratio 2.5, length/beam ratio 7. From this figure it will be clear that in deep water the drift force in beam waves on the block 0.80 vessel corresponds also reasonably well with the

force on a vertical plate. However, at the very reduced water depth of 1.1 times the draft of the vessel, very high drift forces are measured for wave frequencies near the natural frequency of the roll motion. This may be explained by the much higher roll damping (the damping coefficient is approx. 3 times larger) at this water depth, which means that the roll generated waves are higher and consequently the steady drift force too.

For an approximation of the lateral drift force in deep water in regular waves the following expression may be used:

$$Y_d = \frac{1}{2} \rho_w g \zeta_a^2 \cdot R^2 \cdot L \cdot \sin^2 \alpha$$

in which

ζ_a = amplitude of incident wave

R = drift force coefficient for a vertical plate with draft T . R is a function of the dimensionless wave length kT

α = wave direction

For the determination of the mean drift force or the resistance in head waves reference is made to the method described by Gerritsma and Beukelman (12). This method is based on the determination of the radiated energy by calculating the amplitude of the waves generated by the relative vertical motion between ship and waves. In case of a rather flat bow also the influence of the relative surge motion has to be taken into account as is shown in (13).

The wave drift force on other structures

Up to now only the drift force on ship shaped bodies have been dealt with.

For the drift force on semi submersible

type structures no data are available on this moment. Probably the best way to estimate the drift force is to split up the construction in elements consisting of circular or rectangular cylinders and to estimate the drift force on each element separately. In (14) data are given for the drift force on a restrained vertical cylinder.

Wave drift force in irregular waves

When the drift force on a structure is known as a function of the wave frequency either from calculations or model tests, the lateral mean drift force in irregular waves, described by a particular wave spectrum, can be determined from:

$$\bar{Y}_d = \rho_w g \cdot L \cdot \int_0^\infty S_\zeta(\omega) \cdot \frac{Y_d(\omega)}{\frac{1}{2} \rho g \zeta_a^2} d\omega$$

As an empirical approximation for ship shaped bodies the following expression may be used for an irregular sea described by a narrow spectrum:

$$\bar{Y}_d = \frac{1}{16} \rho_w g \zeta_w^2 \frac{1}{3} \cdot R^2(\bar{\omega}) \cdot L \sin^2 \alpha$$

in which

$\zeta_w \frac{1}{3}$ = significant wave height (crest-trough)

$R(\bar{\omega})$ = drift force coefficient for flat plate for $\omega = \bar{\omega}$ = mean wave frequency

Design of the anchor system

If the steady force on a structure in a particular sea state is known and the lay-out of the anchor system has been selected the minimum thickness of the anchor lines is determined by the mean external force as will be illustrated below.

Suppose the anchor system has to satisfy the following criteria

1. The maximum allowable excursion $x_{\max}$ of the structure from its initial unloaded equilibrium position is given as a certain percentage of the water depth.
2. The maximum allowable tension in the anchor lines may not exceed a certain percentage of the breaking strength.

The thickness of the anchor chains is proportional to the weight per unit length (specific weight). The minimum weight required will be attained when the pretension in the anchor lines is such that at an excursion which equals the maximum allowable excursion also the maximum allowable load in the heaviest loaded anchor line is attained.

From the non-dimensional catenary characteristics the tension in the line T_0 divided by $w \cdot h_a$ and the angle θ between the line and the horizontal plane, both measured at the attachment point of the line to the structure, can be determined as a function of the excursion x/h_a of the structure.

w = submerged weight of anchor line per unit length

h_a = height of attachment point above bottom

Since the breaking strength of a particular type of anchor line is proportional to the specific weight w , the excursion of the structure can be determined at which the tension $T_0/w \cdot h_a$ equals the maximum allowable tension. Then the required pretension angle θ can be read off at an excursion which is $x_{\max}$ smaller. The usual non-linear relationship between the excursion of the structure and the horizontal load $F_H/w \cdot h_a$ required for that excursion can be calculated for the

selected pretension.

Subtracting the maximum expected fluctuating motion from the maximum allowable excursion $x_{\max}$ gives the excursion of the structure which can be allowed as a result of the mean force $\bar{F}$ due to wind, waves and current. The total horizontal force $F_H' = F_H/w \cdot h_a$ at this excursion can be read of.

Then the minimum required submerged weight of the anchor lines has to satisfy

$$w = \frac{\bar{F}}{F_H' \cdot h_a}$$

The above described procedure will be illustrated by means of an example.

Example

Question Determine the minimum weight of the anchor lines of an anchoring system for a turret mooring of a drilling vessel having the following main dimensions:

length = 150 m

beam = 26 m

draft = 8 m

wind exposed transverse area = 750 m²

displacement = 17500 metric tons

water depth = 70 m

The design has to be based on a sea state described by a Pierson-Moskowitz spectrum with a significant wave height of 4.5 m and a mean period of 8 sec. The maximum wind speed is 40 knots, the current speed 2 knots. The maximum allowable excursion is 9% of the water depth. The forces in the lines may not exceed 50% of the breaking strength. Although the vessel is equipped with bow and stern thrusters to control the heading, since the turret is located amidships, the anchor system has to be designed for the combined action

of beam wave, wind and current. The system consists of 8 anchor legs equally distributed over the circumference of the turret. From model test data on similar ships the maximum oscillatory excursion of the vessel is estimated to be approximately 4.35 m being 7% of the height h_a of the attachment points of the anchor chains above the bottom.

Solution

The mean force on the vessel determined according the data given in this paper are as follows:

due to a 40 knots beam wind	17 ton
due to a 2 knots beam current	49 ton
due to 4.5 m significant height waves	48 ton
Total	114 ton

From the characteristics of the catenary it can be determined that the anchor legs have to consist of approx. 500 m (8 x h) anchor chain (U-3 quality) or 1050 m (17 x h) steelwire (h = water depth-draft = 62 m) in order to be sure that the tangent of the line at the anchor coincides with the bottom at a tension which equals half the breaking strength.

The pretension angle θ , required to obtain a tension which equals half the breaking strength at the maximum allowable excursion of 9% of the height h_a , amounts to the following values

for 500 m U-3 quality stud link chain:
 $\theta_{\text{pret.}} = 26.2$ degrees
 for 1050 m steelwire: $\theta_{\text{pret.}} = 10.7$ degrees.

The non-dimensional relationships between the excursion of the vessel and the horizontal force required have been calculated for both types of anchor lines. For U-3 stud link chain the result is

shown in Fig. 6. At an excursion of 9% of the height h_a , the tension in the heaviest loaded anchor line attains half the breaking strength T_{obr}

for U-3 quality chain $T_{obr} = 4000 \times w$
for steelwire $T_{obr} = 17500 \times w$

T_{obr} in kg
 w in kg.m^{-1}

Subtracting the maximum expected oscillatory excursion from the maximum allowable excursion leads to an excursion of 2% of the height h_a , that may be allowed as a result of the mean force of 108 ton on the vessel. The dimensionless horizontal load on the system corresponding to this 2% excursion amounts to:

for U-3 chain 8.7

for steelwire 59.1

The resulting minimum required submerged weight of the anchor lines is given in Table VI. The corresponding approximate diameters of the lines are also mentioned in Table VI.

The method described here has to be adapted for each special case. However, the example illustrates clearly the important role which the mean force may play for the determination of the anchor system.

Nomenclature

a_n, b_n Fourier coefficients
 h depth of water
 h_a height of attachment point of anchor line above bottom
 k wave number = $2\pi/\lambda$
 p pressure
 p_0 atmospheric pressure
 t time
 w submerged weight of anchor line
 x excursion

x_{max} maximum allowable excursion of structure
 x, y, z right handed system of coordinates
 z vertical coordinate, upward positive
 A_L lateral area above water surface
 A_{LS} submerged lateral area
 A_T transverse area above water surface
 A_{TS} submerged transverse area
 B breadth of ship
 C_{nw} yaw wind moment coefficient
 C_{nc} yaw current moment coefficient
 C_{xw} longitudinal wind force coefficient
 C_{xc} longitudinal current force coefficient
 C_{yw} transverse wind force coefficient
 C_{yc} transverse current force coefficient
 $\bar{F}$ total mean load on anchored structure
 F_H horizontal load on anchor system
 N_C yaw current moment
 N_W yaw wind moment
 R non-dimensional drift force coefficient
 T draft of ship
 T_O tension in anchor line
 V velocity of water motion
 V_w wind velocity
 V_c current velocity
 X_C longitudinal current force
 X_W longitudinal wind force
 Y_c transverse current force
 Y_d transverse drift force
 $\bar{Y}_d$ mean transverse drift force in irregular waves

Y_w	mean transverse wind force
α	angle of incidence
ζ_a	amplitude of incident wave
ζ_{ar}	amplitude of reflected and scattered wave
ζ_w	significant wave height
θ	angle between anchor line and horizontal plane at the attachment point to the structure
θ_{pret}	pretension angle = angle θ for T_0 is equal to pretension
ϕ	velocity potential
ρ_a	density of air
ρ_w	density of water
ω	wave frequency
$\bar{\omega}$	mean wave frequency of an irregular sea

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TABLE 2 - DATA OF TANKERS

Ship No.	Type	Length	Condition	Data taken from ref.
1	bridge amidsh.	-	loaded	2
2	" "	-	ballast	2
3	bridge aft	-	loaded	2
4	" "	-	ballast	2
5	bridge amidsh.	225 m.	loaded	3
6	" "	"	ballast	3
7	bridge aft	"	loaded	3
8	" "	"	ballast	3
9	" "	172 m.	loaded	4
10	" "	150 m.	loaded	5
11	" "	"	ballast	5

TABLE 3 - TRANSVERSE FOR THE STATISTICAL FORCE OF TANKERS DUE TO WIND

Ship No.	a_0	a_1	a_2	a_3	a_4	a_5
1	-0.131	0.733	-0.055	0.059	0.133	-0.001
2	-0.079	0.015	-0.194	0.133	0.078	0.025
3	-0.028	0.799	-0.077	-0.054	0.018	-0.013
4	0.014	0.732	-0.055	-0.017	-0.018	-0.053
5	-0.074	1.050	0.017	-0.062	0.080	-0.110
6	-0.055	0.748	0.018	-0.013	0.015	-0.151
7	-0.035	0.030	0.031	0.012	0.021	-0.072
8	-0.033	0.046	-0.034	0.024	-0.031	-0.039
9	-0.042	0.467	-0.072	0.109	0.075	-0.017
10	-0.075	0.711	-0.062	0.043	0.044	-0.036
11	-0.051	0.577	-0.038	0.051	0.042	0.006

TABLE 3 - COEFFICIENTS FOR THE TRANSVERSE FORCE ON TANKERS DUE TO WIND

Ship No.	b_1	b_2	b_3	b_4	b_5
1	0.765	0.039	0.003	0.034	-0.019
2	0.630	0.004	0.002	-0.004	-0.003
3	0.697	0.035	0.018	0.026	-0.023
4	0.785	0.014	0.014	0.015	-0.020
5	0.797	-0.013	0.023	0.007	-0.044
6	0.731	-0.014	0.016	0.001	-0.025
7	0.718	0.032	0.010	-0.001	-0.040
8	0.735	0.003	0.004	-0.005	-0.017
9	0.764	0.037	0.052	0.016	-0.003
10	0.619	0.051	0.023	0.032	-0.032
11	0.879	0.026	0.014	0.031	-0.020

TABLE 4 - COEFFICIENTS FOR THE YAW MOMENT ON TANKERS DUE TO WIND

Ship No.	b_1	b_2	b_3	b_4	b_5
1	-0.0451	-0.0617	-0.0110	-0.0110	-0.0010
2	-0.0338	-0.0600	-0.0060	-0.0096	-0.0013
3	-0.0765	-0.0571	-0.0166	-0.0146	0.0021
4	-0.0524	-0.0738	-0.0175	-0.0089	-0.0021
5	-0.0216	-0.0531	-0.0063	-0.0073	0.0024
6	-0.0059	-0.0730	-0.0035	-0.0017	-0.0013
7	-0.0526	-0.0596	-0.0111	-0.0112	0.0099
8	-0.0335	-0.0722	-0.0090	-0.0047	0.0067
9	-0.1025	-0.0721	-0.0345	-0.0127	-0.0022
10	-0.0381	-0.0681	-0.0202	-0.0145	0.0030
11	-0.0644	-0.0726	-0.0244	-0.0076	0.0024

TABLE 5 - COEFFICIENTS FOR THE TRANSVERSE FORCE AT YAW MOMENT ON TANKERS DUE TO CURRENT FORCE FOR THE 100 FT. CONDITION IN DEEP WATER

	TRANSVERSE FORCE	YAW MOMENT
b_1	0.008	-0.0007
b_2	0.000	-0.0014
b_3	-0.116	0.0009
b_4	0.000	0.0010
b_5	-0.043	0.0011

TABLE 6 - THE MINIMUM REQUIRED STRENGTH WIRE AND THE APPROXIMATE TRANSDUCER ALPHAS LINE

	submerged weight	diameter
	mm	mm
U-3 steel link chain	241 kg	107 mm # 4 1/4"
steel wire	31.7 kg	22 mm # 3 5/8"

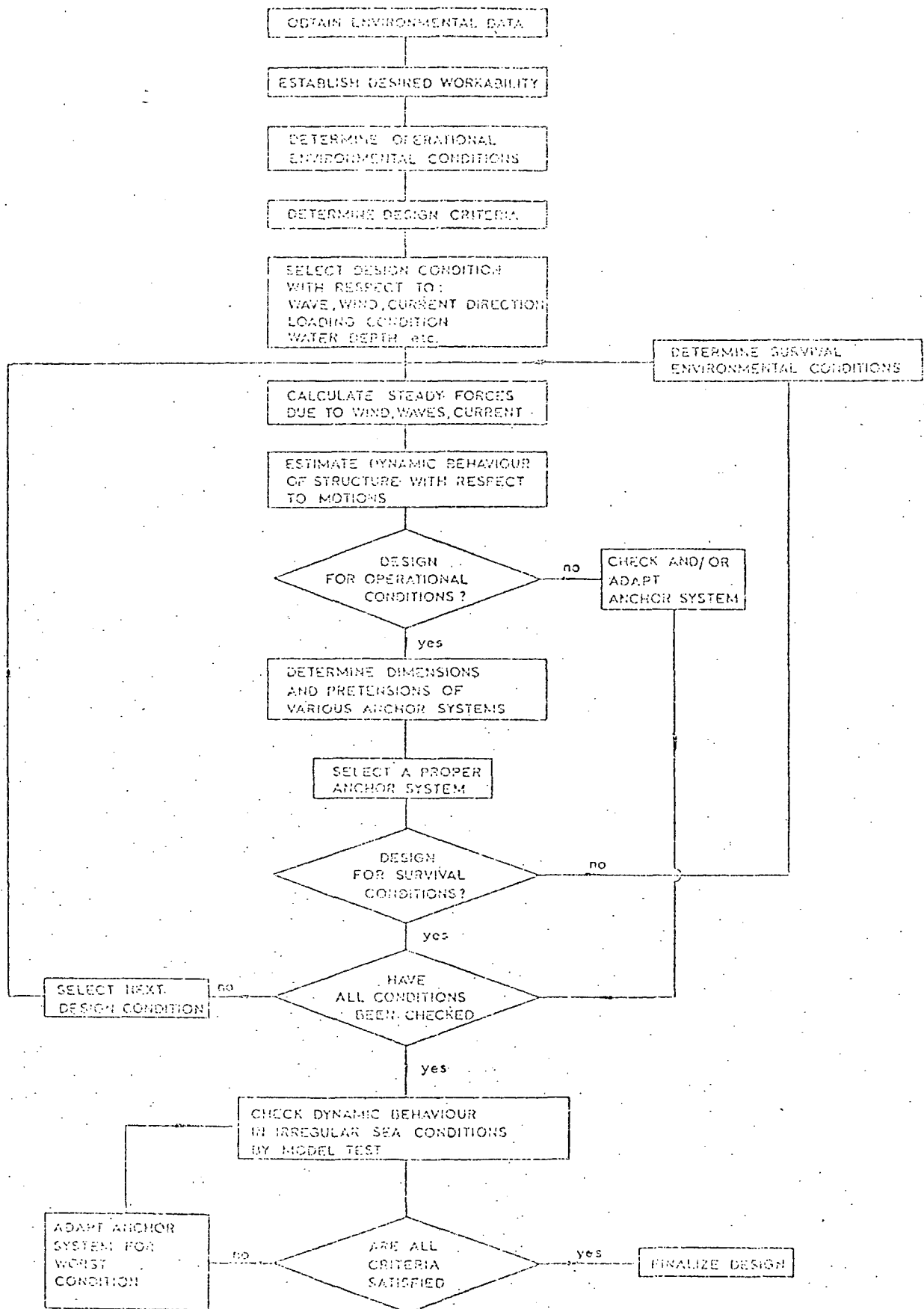


Fig. 1 - The design of an anchor system.

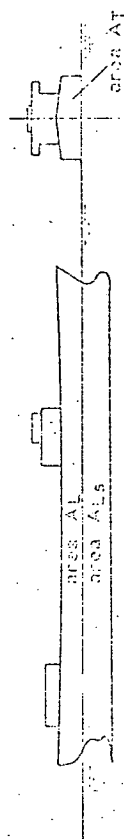
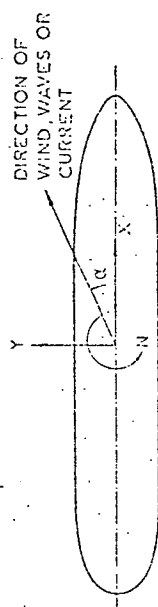


FIG. 2 - Representation of forces and moments.

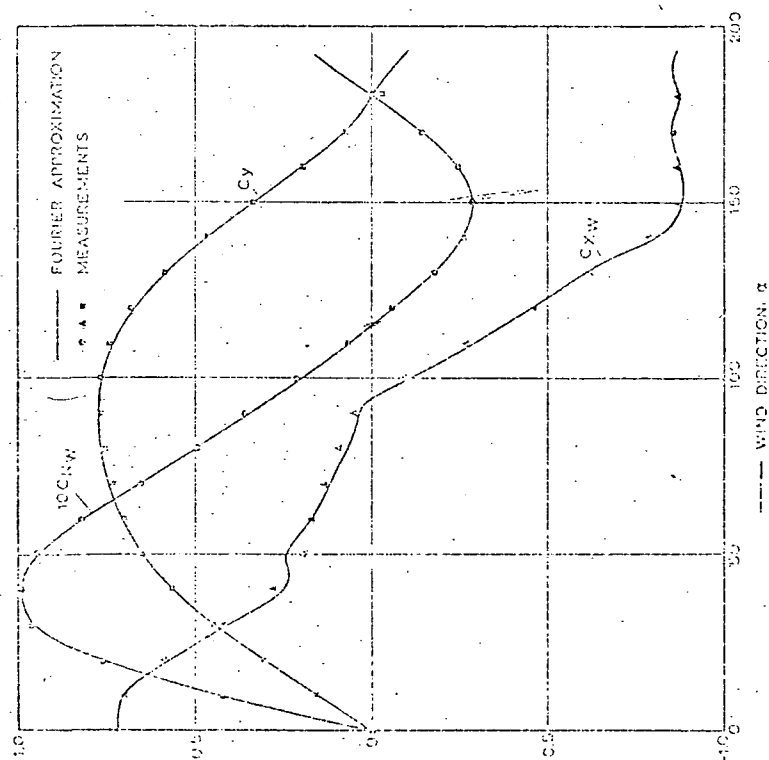


FIG. 3 - Comparison of the Fourier approximation with measurements of the wind load on the hull.

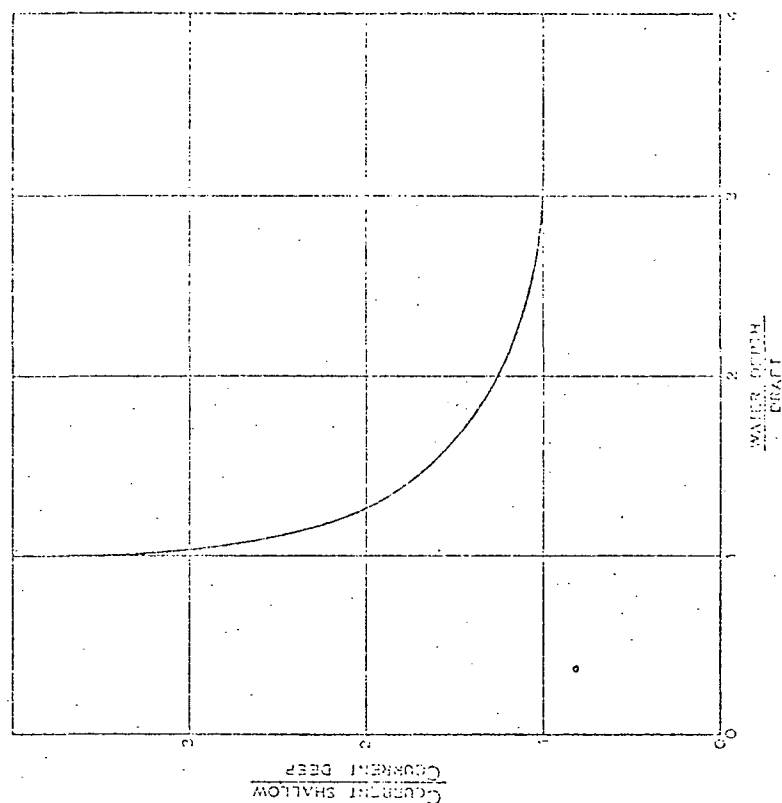


FIG. 4 - The influence of the water depth on the current for a fixed α .

DRIFT FORCE COEFFICIENT $R = \sqrt{\frac{Y_d}{\frac{1}{2} \rho_w g L^2 L}}$
DEEP WATER

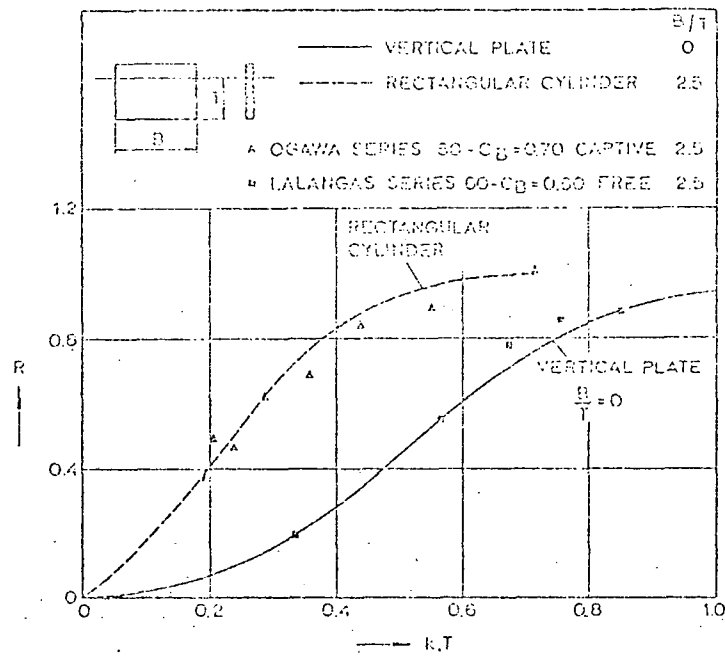


Fig. 5 - Drift force coefficient in beam waves.

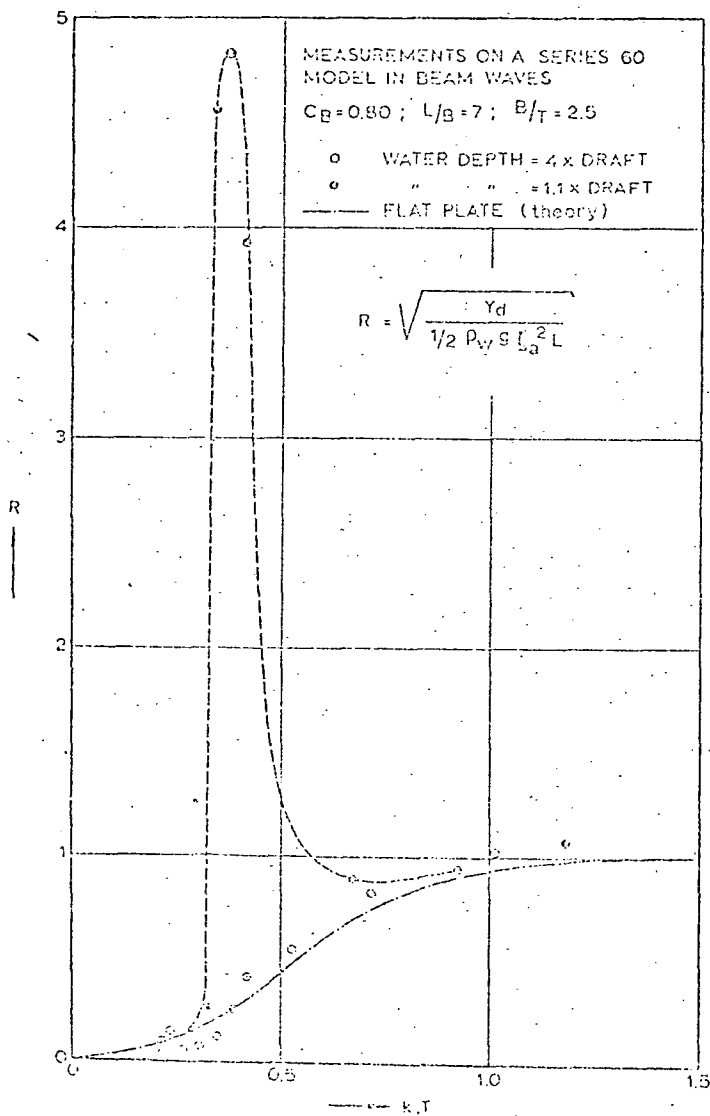


Fig. 6 - Influence of water depth on the drift force coefficient.

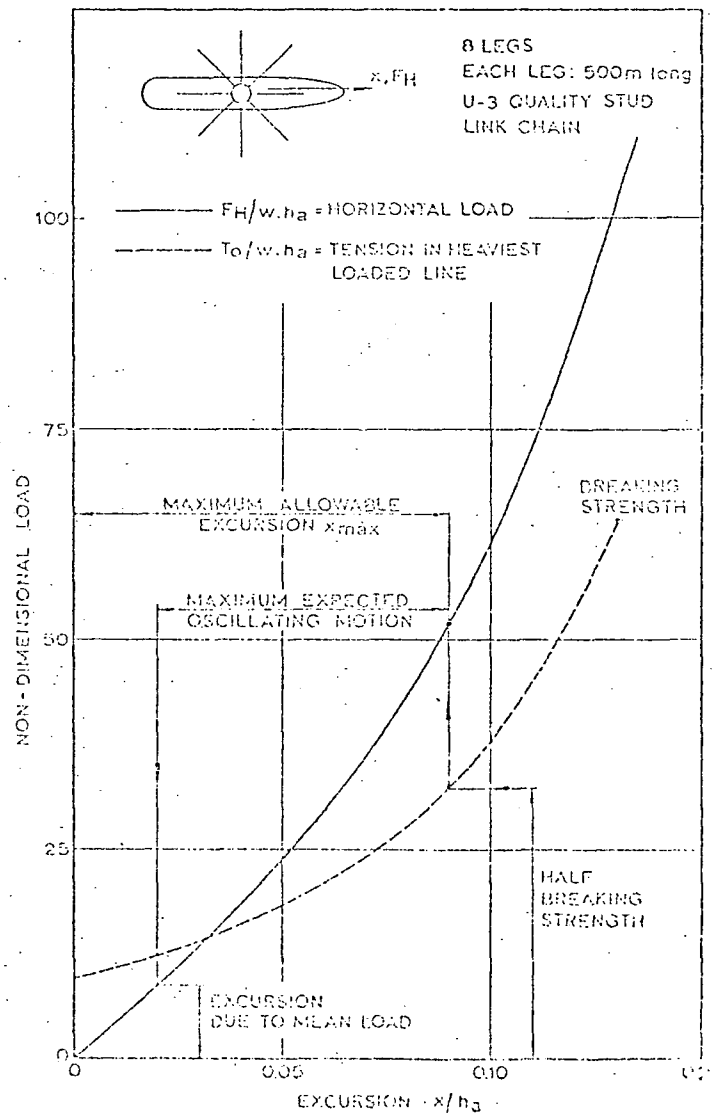


Fig. 7 - Characteristics of the eight-leg mooring system.